

Macro and Option Information in Systematic FX Carry

Predicting the distribution of carry returns, and turning that prediction into exposure

Theo Li

Week 6 Research Update

1 Executive summary

This study asks whether macroeconomic state variables and option-implied information can improve a systematic FX carry portfolio, and — more precisely — *which moment* of the carry return distribution each source of information actually predicts. The sample is a 20-currency pooled G10 and EM panel of 234 monthly observations running from January 2007 to June 2026, split chronologically into a 140-month discovery window, a 47-month validation window and a 47-month frozen evaluation window.

Across 25 point-in-time macro features, predictive evidence is broadest for **carry tail risk**: eight features carry supported evidence on the severe-loss indicator, four on return magnitude, and three on the mean return. Macro information is therefore not silent about expected returns, but the weight of the evidence — in breadth, in cross-sample consistency, and in economic coherence — sits with adverse outcomes rather than with average ones. That asymmetry is what motivates using macro information as a defensive gross-exposure overlay rather than as a directional timing model.

The oil overlay is the cleanest illustration. Halving gross exposure when the one-month oil return sits in the lower tail of its own rolling distribution raises the full-sample gross Sharpe from 0.578 to 0.651 and improves expected shortfall by 1.17 percentage points. Decomposed conditionally, the rule earns an annualised +16.30% relative return in months where the signal was active *and* baseline carry subsequently lost money, and pays back −12.03% on the same basis in active months where baseline gained. At 41.3% event precision the two effects almost exactly cancel, netting −0.06% annualised across all active months. The Sharpe improvement is therefore a variance result, not an alpha result.

Option-derived information improves portfolio construction through two mechanisms that must be kept apart. Selection-based conditioning (architecture A+B) reaches a full-sample Sharpe of 0.688 at unchanged average gross exposure of 2.00, while variable-gross option-risk scaling (architecture A+C) reaches 0.699 at average gross of 1.14 with a maximum drawdown of −11.6% against baseline's −19.0%. Reporting the two without also reporting average gross would credit exposure reduction as if it were selection skill.

Combining the two information sets is evaluated *against the option parent*, not against baseline. Of 200 parent-feature combinations, 89 raise the parent's Sharpe and 111 lower it, and the improvements concentrate in the same EM-equity, commodity and oil features that the regressions had already flagged. Parents that already manage their own exposure gain least.

Finally, none of this survives a strict statistical screen. Of 270 active non-reference strategies, 36 have a raw HAC *p*-value below 0.05 on their mean active return and **none** has a Benjamini-Hochberg *q*-value below 0.10. The Tier 1 robust shortlist is empty. The defensible contribution of this work is a claim about mechanism — which moment macro information predicts, and which portfolio action that justifies — not a validated tradable edge.

Chronological, non-overlapping samples: 60 / 20 / 20 per cent of 234 months

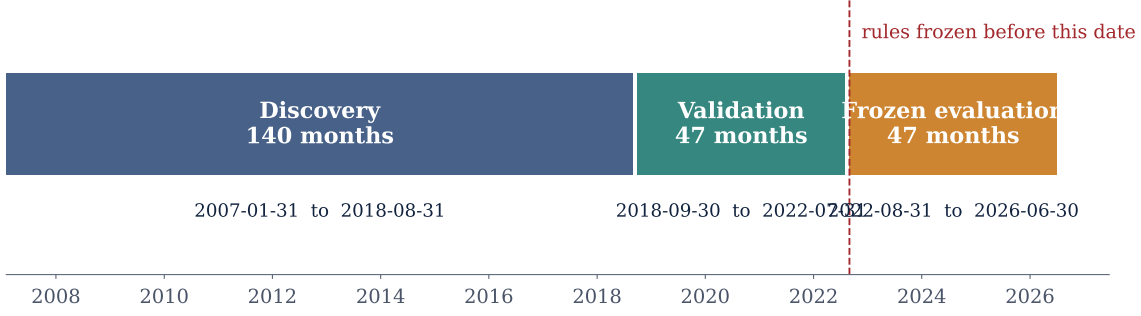


Figure 1. Chronological, non-overlapping sample design. Evidence labels are assigned using discovery and validation only; the evaluation window is frozen and used solely as robustness evidence.

2 Research motivation and questions

A carry portfolio that is long high-yielding currencies against low-yielding funders earns a persistent premium and periodically returns several years of gains inside a few months. Over 2007–2026 the pooled baseline used here earns an annualised 5.24% at 9.07% volatility, a gross Sharpe of 0.578, with a maximum drawdown of -19.0% . The losses are not uniformly distributed through time; they cluster in identifiable risk-off states. That clustering is what makes conditioning information potentially valuable, and it also determines what kind of conditioning is worth looking for.

The study is organised around three questions.

- RQ1** Do formation-time macro conditions predict the *mean*, the *magnitude*, or the *tail* of next-month carry returns? These are three separate questions with three different answers, and conflating them is the most common way to over-read macro evidence.
- RQ2** Can a macro risk signal be converted into a defensive gross-exposure rule that improves risk-adjusted performance, without becoming a directional timing model?
- RQ3** Do macro signals add incremental information *after* option-derived conditioning, judged against the option-conditioned parent rather than against baseline carry?

The distinction that runs through all three is between forecasting the sign of a return and forecasting its size. A signal that fails at the first can still succeed at the second. The first justifies changing the cross-sectional ranking; the second justifies changing the scale of the book. Most of the results below turn on keeping those two claims apart.

3 Data and sample construction

The return panel is the pooled G10 and EM carry book inherited from the earlier notebooks in this project: 20 currencies ranked jointly on carry, with a top basket held long and a bottom basket held short at equal weights within each leg. Monthly portfolio returns, weights, turnover and cost series are reproduced inside Notebook 08 from the stored Notebook 07 weight files and verified to agree with the stored performance outputs to machine precision (maximum absolute difference 0.0 across all reconstructed core streams).

The sample runs from 2007-01-31 to 2026-06-30, 234 months, split as follows: discovery 2007-01-31 to 2018-08-31 (140 months), validation 2018-09-30 to 2022-07-31 (47 months), and frozen evaluation 2022-08-31 to 2026-06-30 (47 months). The split is chronological and non-overlapping. This is not a stylistic preference: macro states, implied volatility and carry returns are all strongly persistent, so a random train/test partition would place neighbouring months on opposite sides of the boundary and allow a rule to learn from its own immediate future. A chronological split also matches the way the strategies would have been deployed — rules fixed at a point in time, then applied forward.

Table 1. The macro universe by economic category. Full feature-level detail, including risk signs and evidence flags, is in Appendix A.

Category	Series	#	Transforms	Risky	Economic reading
EM equity	MXEF	4	$r1m$, $mom3m$, $dd6m$, $r1m(tail)$	low	EM equity
commodity	COMMODITY	2	$r1m$, $mom3m$	low	broad commodity index
dollar	DXY	2	$r1m$, $mom3m$	high	broad dollar
equity	SPX	4	$r1m$, $mom3m$, $dd6m$, $r1m(tail)$	low	US equity
oil	OIL	2	$r1m$, $mom3m$	low	front WTI
rates	CURVE	2	level, $\Delta1m$	high	US 2s10s slope
rates	US10Y	1	$\Delta1m$	high	long US yield
rates	US2Y	1	$\Delta1m$	high	front-end US yield
rates vol	MOVE	2	level, $\Delta1m$	high	US rates volatility
volatility	JPMVXYEM	2	level, $\Delta1m$	high	EM FX implied volatility
volatility	VIX	3	level, $\Delta1m$, $\Delta3m$	high	US equity-implied volatility

Only discovery and validation are used to assign the evidence labels reported in Section 6. The evaluation window is quoted as robustness evidence and never as the descriptive headline; at 47 months the standard error of an annualised Sharpe estimate is roughly $\sqrt{12/47} \approx 0.5$, which is larger than any Sharpe difference reported in this study.

4 Macro feature engineering

4.1 The macro universe

Eleven market series are used, grouped by the economic role they play rather than by data vendor. Implied volatility measures the *price* of risk: VIX for US equity, JPMVXYEM for EM FX — the volatility of the carry basket itself — and MOVE for US rates. Equity and commodity returns measure *realised* risk: SPX, MXEF, a broad commodity index and front WTI. The dollar and the US curve measure *transmission*: DXY, the 2-year and 10-year US yields, and the 2s10s slope.

The organising idea is that these are not eleven independent bets on the world. They are four readings of one underlying state: how much risk is being priced, and how much of it has already been realised. This matters later, when composite signals built from many features turn out to add less than their apparent breadth would suggest. MXEF and oil deserve particular attention because the long leg of the book is predominantly EM: those two series sit closest to the exposures the portfolio actually carries.

4.2 Transformations

Each series is converted into one or more formation-time features, evaluated at the month end at which the portfolio is formed:

- **Levels** for series whose absolute value is economically meaningful — VIX, MOVE, JPMVXYEM, the 2s10s slope.
- **Ordinary differences** over one and three months for yields and volatility indices, where the change in the level rather than the level carries the news.
- **One-month log returns** for price series: $\log(P_t/P_{t-1})$.
- **Three-month log momentum**: $\log(P_t/P_{t-3})$, a slower reading of the same channel.
- **Six-month drawdown** for equity indices, $P_t / \max_{s \in [t-5, t]} P_s - 1$, which is zero at a new high and becomes increasingly negative as the index falls away from its recent peak.
- **Tail-return labels** for SPX and MXEF, intended to isolate the lower tail of the one-month return distribution.

This produces 25 features from 11 series. Two of them — `SPX_tail_return_1m` and `MXEF_tail_return_1m` — are numerically identical to the corresponding `_return_1m` features: the label describes the intended reading, not a different transformation. The notebook flags both as

duplicate-equivalent and excludes them from the paired lead-versus-contemporaneous comparison, while retaining them in the evidence tables. This is stated explicitly here because treating them as independent findings would overstate the breadth of the mean-return and tail-risk results in Section 6.

4.3 Historical normalisation

Each raw feature $X_{j,t}$ is converted to a historical z -score

$$z_{j,t} = \frac{X_{j,t} - \bar{X}_{j,[t-60, t-1]}}{s_{j,[t-60, t-1]}}, \quad (1)$$

where $\bar{X}$ and s are the mean and standard deviation over a trailing 60-month window requiring at least 24 observations, both lagged one month. Equation (1) is the point-in-time core of the whole macro layer: the statistic evaluated at t uses only data observable strictly before t . A feature is therefore measured as unusual relative to its own recent history, not relative to a full-sample distribution that would not have been known at the time.

4.4 Risk direction

Features differ in the direction that constitutes risk. A high VIX reading is risky; a *low* oil return is risky. To place all features on a common scale the notebook applies a risk-sign multiplier,

$$\tilde{z}_{j,t} = \rho_j z_{j,t}, \quad \rho_j = \begin{cases} +1 & \text{higher is riskier,} \\ -1 & \text{lower is riskier,} \end{cases} \quad (2)$$

with 13 features classified as higher-is-riskier (volatility indices, the dollar, yields, the curve) and 12 as lower-is-riskier (equity, EM equity, commodity, oil). After the transformation, a large $\tilde{z}_{j,t}$ always means more risk regardless of the underlying series.

The convention buys three things. It allows a single event-threshold rule to cover all 25 features rather than requiring an upper threshold for some and a lower threshold for others. It gives every regression a single expected sign, so that a coefficient with the wrong sign is unambiguously evidence against the hypothesis. And it makes averaging across features coherent, which is what the composite signal in Section 8.5 relies on.

4.5 Point-in-time construction

Three quantities move together through time and all three are lagged: the normalising mean, the normalising standard deviation, and the rolling percentile threshold that later defines a risk event. The notebook runs lineage audits over the macro features, the option signals and the strategy-level return timing; all report **zero** future-horizon violations, meaning no signal used to form a position at month t draws on information dated after t . Section 11 records the limits of that statement, which is a claim about detected violations rather than a proof of correctness.

5 Macro regression framework

5.1 Predictive and contemporaneous specifications

All macro regressions are *univariate*: one macro predictor at a time, with an intercept. There is no 25-variable multivariate regression anywhere in the analysis, and the counts of “supported” features reported below should not be read as if they came from one.

Two timings are estimated. The predictive regression

$$Y_{t+1} = \alpha + \beta X_t + \varepsilon_{t+1} \quad (3)$$

relates a formation-time macro reading to an outcome realised over the following month. The contemporaneous regression

$$Y_{t,t+1} = \alpha + \beta X_{t,t+1} + \varepsilon_{t,t+1} \quad (4)$$

relates the outcome to the macro move that occurs *during* the holding period. Equation (3) is a candidate signal. Equation (4) is a description of a mechanism and can never be traded, because $X_{t,t+1}$ is not known when the position is formed. The two are reported side by side because the gap between them is informative, but they are never mixed.

5.2 The regression grid

Each regression is indexed by six dimensions: macro feature (25); predictor form (continuous standardised z -score, or binary macro risk-event indicator); timing (predictive or contemporaneous); portfolio component (full book, long leg, short-position leg, or the currency-level panel); outcome; and sample (full, discovery, validation, evaluation). This is a pre-defined grid, not a search: every combination is estimated and reported, including the ones that fail.

5.3 Outcome families

Three families of dependent variable separate the three questions in RQ1:

$$\underbrace{r_{t+1}}_{\text{mean}}, \quad \underbrace{|r_{t+1}|, r_{t+1}^2}_{\text{magnitude}}, \quad \underbrace{\mathbf{1}\{r_{t+1} < 0\}, \mathbf{1}\{r_{t+1} \leq q_t^{5\%}\}}_{\text{tail}}. \quad (5)$$

Here $r_{t+1} = \sum_i w_{i,t} R_{i,t+1}$ is the gross portfolio return; long-leg and short-position contributions are the same sum restricted to positive and negative weights respectively. A severe loss is a month falling in the lower 5% of the book's *own* trailing 60-month return distribution, with the threshold lagged one month exactly like the normalising moments in Equation (1).

5.4 Inference

Estimation is OLS with a HAC/Newey–West covariance matrix using three lags and a minimum of 18 observations; currency-panel variants cluster by month, require at least 80 observations, and include currency fixed effects together with EM and carry-rank interactions. Overlapping information and persistent regressors make the HAC correction necessary rather than cosmetic. The notebook records diagnostics explicitly rather than silently dropping cases: 36 regressions are skipped for insufficient variation and two produce an invalid covariance matrix and are flagged as *invalid inference* rather than reported as null results.

5.5 Multiple testing

Benjamini–Hochberg and Holm adjustments are applied within each (timing, sample, dependent-variable) family, and nominal p -values are always displayed beside their adjusted q -values. No individual nominal p -value is treated as decisive on its own. The same discipline is applied at the strategy level in Section 11, where it has considerably sharper consequences.

5.6 Standardised coefficients

Coefficients are reported as *standardised economic effects*: the change in the outcome associated with a +1 regression-sample standard deviation move in the predictor, expressed in basis points for return outcomes and percentage points for probability outcomes. This makes effect sizes comparable across macro variables *within* a single dependent variable. It does not make them comparable across dependent variables, and no comparison of that kind is made anywhere in this document. Event-indicator regressions are not standardised — a binary predictor has no natural scale — and are reported as an active-minus-inactive difference in means. A standardisation invariance audit passes on all 1,880 continuous regressions.

6 Macro predictive evidence

6.1 What counts as supported

A feature is labelled *supported* for an outcome family when it shows the economically expected sign in both the discovery and the validation sample, has valid inference in both, and reaches clear significance in discovery ($p < 0.10$ or BH $q < 0.20$). *Directionally suggestive* means the

Table 2. Supported predictive features by outcome family, with full-sample standardised coefficients. Coefficients are comparable within a block and not across blocks. Source: 08v6_predictive_portfolio_regressions.parquet and 08v6_predictive_evidence_map.parquet.

Outcome	Feature	Std. coef.	HAC t	HAC p	BH q
Mean return	MXEF_return_1m	+31.44	+1.51	0.130	0.835
	MXEF_tail_return_1m	+31.44	+1.51	0.130	0.835
	COMMODITY_momentum_3m	+24.76	+1.67	0.094	0.835
Risk magnitude	VIX_level	+39.66	+2.95	0.003	0.027
	JPMVXYEM_change_1m	+24.52	+2.61	0.009	0.065
	MXEF_momentum_3m	-41.77	-3.39	0.001	0.018
	MXEF_drawdown_6m	-41.73	-3.61	0.000	0.015
Tail risk	JPMVXYEM_change_1m	+1.27	+1.07	0.285	0.619
	MXEF_return_1m	-2.55	-2.37	0.018	0.147
	MXEF_momentum_3m	-2.79	-2.49	0.013	0.147
	MXEF_drawdown_6m	-3.02	-2.42	0.015	0.147
	MXEF_tail_return_1m	-2.55	-2.37	0.018	0.147
	COMMODITY_return_1m	-3.33	-2.10	0.036	0.224
	COMMODITY_momentum_3m	-2.52	-2.12	0.034	0.224
	OIL_return_1m	-2.75	-1.89	0.058	0.266

expected sign without the significance condition. The frozen evaluation window plays no part in the label. Given the risk-sign convention, the expected continuous coefficient is $-\rho_j$ for the mean outcome and $+\rho_j$ for magnitude and tail: more risk should mean a lower expected return, larger absolute moves and a higher severe-loss probability.

6.2 Results by outcome family

Mean returns. Three features are supported: MXEF_return_1m, MXEF_tail_return_1m and COMMODITY_momentum_3m. All three carry the expected positive sign after risk-direction normalisation — weak EM equity and weak commodity momentum at formation are followed by lower carry returns. The standardised effects are economically non-trivial (roughly +25 to +31 basis points per standard deviation on the monthly portfolio return) but the full-sample HAC p -values are 0.13, 0.13 and 0.09, and the BH q -values are 0.835 across the board. Two of the three are also the same underlying series, and one of them is the duplicate-equivalent tail label. The correct reading is that mean-return predictability is present and narrow, not that it is absent — and not that it is established.

Risk magnitude. Four features are supported: VIX_level, JPMVXYEM_change_1m, MXEF_momentum_3m and MXEF_drawdown_6m. These are considerably stronger statistically, with HAC p -values between 0.000 and 0.009 and BH q -values between 0.015 and 0.065. A high VIX level at formation is associated with roughly +40 basis points of additional absolute portfolio return over the following month per standard deviation; weak EM equity momentum works in the same direction. This is the family where a formation-time reading of priced risk translates most reliably into a statement about the size of the coming month’s move.

Tail risk. Eight features are supported — JPMVXYEM_change_1m, four MXEF transformations, both commodity features and OIL_return_1m — making this both the broadest family and the most economically coherent one. The supported set is dominated by the EM-equity, commodity and oil complex, which is precisely the group most directly connected to the long leg of the book. Standardised effects run around -2.4 to -3.3 percentage points of severe-loss probability per standard deviation, with HAC p -values from 0.013 to 0.058 and BH q -values from 0.147 to 0.266.

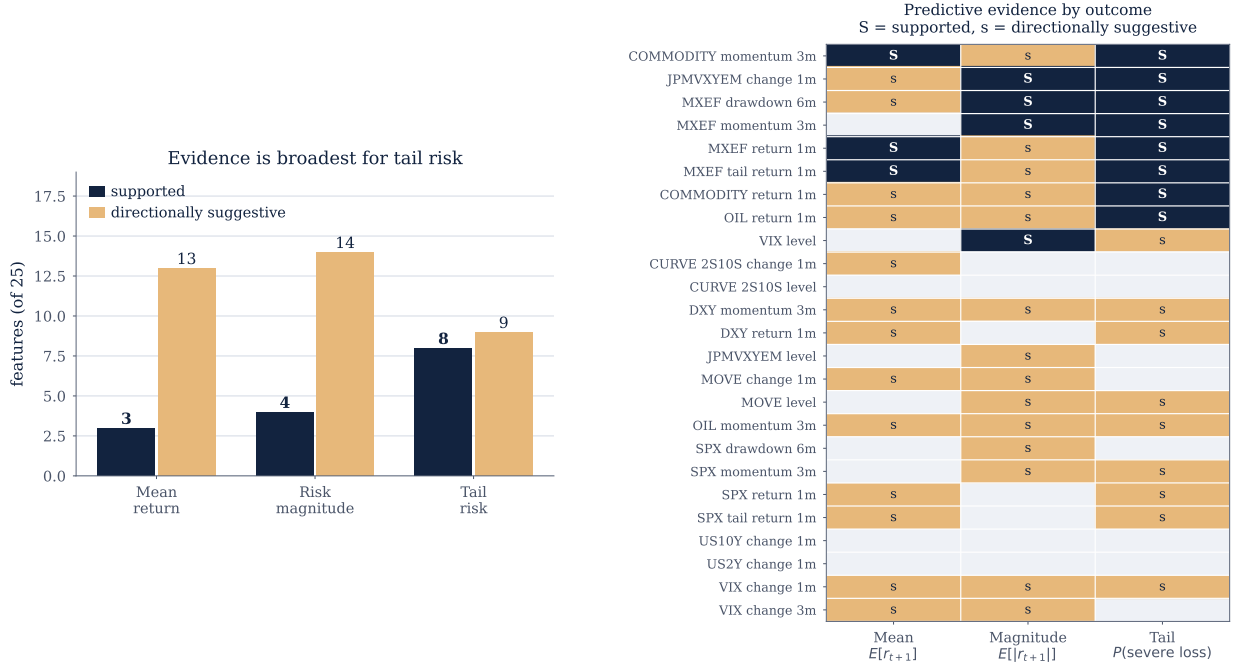


Figure 2. Left: counts of supported and directionally suggestive features by outcome family. Right: the full 25-feature evidence grid, where S marks supported evidence and s directionally suggestive.

The conclusion, stated precisely. Macro information is *not* silent on mean returns: three features clear the support bar. But breadth, cross-sample consistency and economic coherence all sit with adverse tail outcomes, with intermediate evidence for magnitude. Macro predictive evidence is strongest for carry tail risk, with additional evidence for risk magnitude and more limited but non-zero evidence for mean returns. This tells us which moment to trade; it does not tell us that any of it is tradable.

6.3 Lead versus contemporaneous effects

Figure 3 plots the 14 explicitly matched semantic pairs: for each concept, the formation-time predictive effect beside the matched holding-period contemporaneous effect, both in standardised units and therefore directly comparable within a panel. Nine predictive features have no same-concept holding-period counterpart and are excluded from the plot while remaining in the tables; the two duplicate-equivalent tail-return features are also excluded.

The pattern is stark and consistent: contemporaneous effects are almost always far larger than predictive ones, in both the mean-return and severe-loss panels. Macro variables explain a great deal of carry performance *ex post* and comparatively little *ex ante*. That gap is the difference between a mechanism and a signal, and it is the single most important reason the strategy work that follows targets exposure rather than direction: we can identify the states in which carry gets hurt long before we can forecast the return itself. Contemporaneous regressions appear in this document as mechanism evidence only and are never used to construct a trading rule.

7 Macro risk overlay construction

7.1 Event thresholds

A continuous z -score becomes a portfolio rule through a tail-of-own-history event definition:

$$E_{j,t} = \begin{cases} \mathbf{1}\{z_{j,t} \geq Q_{j,[t-60,t-1]}^{80}\}, & \rho_j = +1, \\ \mathbf{1}\{z_{j,t} \leq Q_{j,[t-60,t-1]}^{20}\}, & \rho_j = -1, \end{cases} \quad (6)$$

where Q^{80} and Q^{20} are rolling 60-month percentiles computed on prior observations only, exactly like the moments in Equation (1). The asymmetry between the two branches is entirely handled

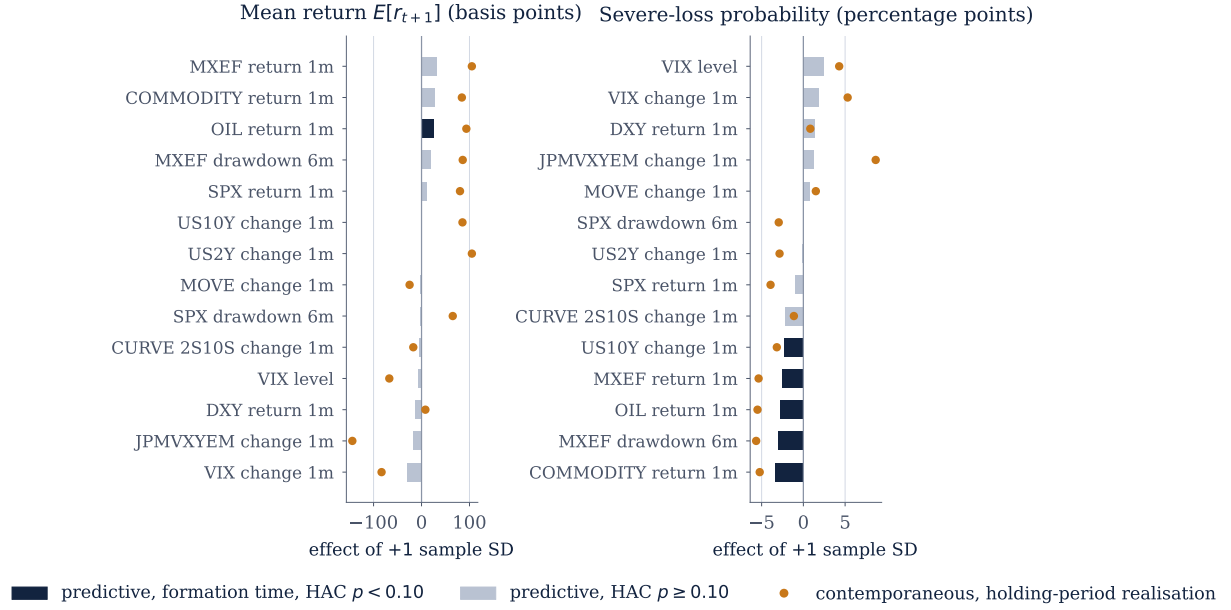


Figure 3. Predictive versus contemporaneous standardised effects for the 14 matched pairs. Bars are formation-time effects; dots are the matched holding-period realisation. Source: `08v6_lead_concurrent_plot_pair_audit.parquet`.

by the risk sign: for a higher-is-riskier feature the upper tail is the risk state, and for a lower-is-riskier feature the lower tail is. Events are undefined before enough history accumulates, and in that case default to the inactive state — no history means no de-risking.

The 80th/20th percentile choice makes events roughly a fifth of all months (46 of 234 for oil, a 19.7% activation rate): frequent enough to matter for long-run performance, rare enough to leave the carry premium substantially intact.

7.2 Gross scaling

When an event fires, every weight in the parent portfolio is multiplied by a single scalar:

$$w_{i,t}^{\text{macro}} = m_t w_{i,t}^{\text{parent}}, \quad m_t = \begin{cases} 0.50 & E_t = 1, \\ 1.00 & E_t = 0. \end{cases} \quad (7)$$

Because a single scalar multiplies all weights, the *composition* of the portfolio is untouched and only its size changes. This is a deliberate design choice, not a simplification: the evidence in Section 6 supports statements about when losses cluster, not about which currencies will outperform, and a rule that reordered the cross-section would be trading a claim the regressions do not support.

The 50% multiplier is deliberately blunt. It is not estimated, optimised or tuned against realised returns, which removes one obvious channel for in-sample fitting at the cost of leaving the rule un-optimised.

7.3 Zero-scaling sensitivity

A 0% variant — flattening the book entirely on an event — is carried through the whole analysis as a sensitivity. It often looks marginally better on Sharpe. It is deliberately *not* the primary rule: taking a portfolio to zero on the reading of a single indicator is a much stronger claim about signal reliability than the evidence in Section 6 supports, and the difference between the two variants is a useful measure of how much of any result depends on the aggressiveness of the response.

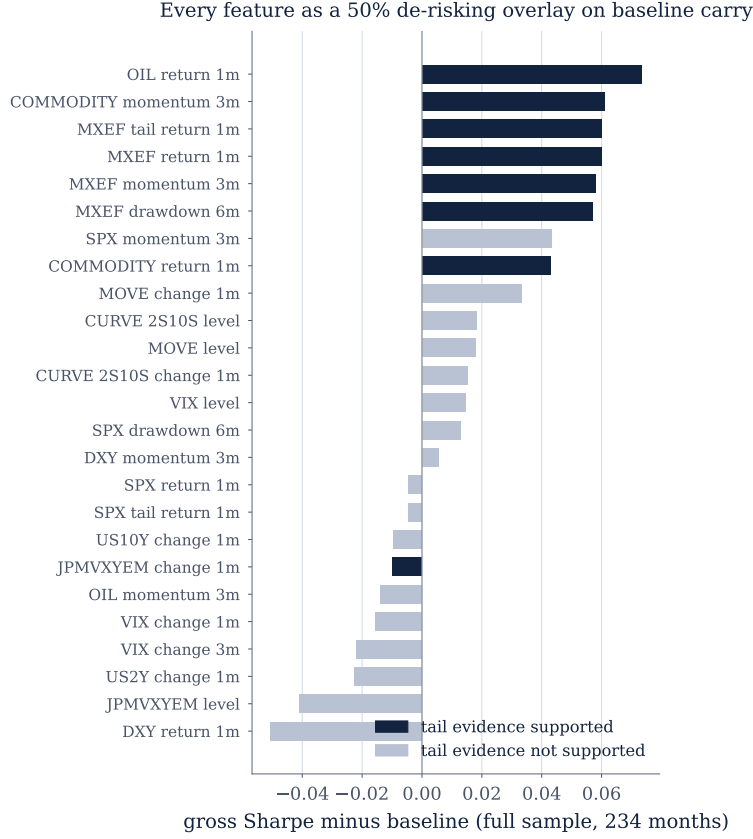


Figure 4. Every macro feature as a 50% de-risking overlay on baseline carry, full sample. Dark bars mark features whose tail evidence was labelled supported before any portfolio was constructed.

7.4 Gross exposure

Gross exposure is

$$\text{Gross}_t = \sum_i |w_{i,t}|, \quad (8)$$

which equals 2.00 for the baseline book by construction (unit gross on each leg, net zero). A 50% event therefore moves the baseline from gross 2.00 to gross 1.00 in active months, and the long-run average gross for the oil overlay is 1.80. For a variable-gross parent, Equation (7) halves whatever the parent was already running rather than forcing gross to any particular level — a distinction that matters in Section 10.

8 Standalone macro strategy results

8.1 The overlay leaderboard

Applying Equations (6) and (7) to baseline carry for each of the 25 features produces 25 strategies that differ only in their trigger. Figure 4 and Table 3 report the outcome.

Three observations matter more than the ranking itself. First, the ordering is not purely a backtest artefact: the features at the top are largely the ones whose *tail* evidence was labelled supported before any portfolio was built, so the regression evidence and the portfolio evidence agree on the EM-equity, commodity and oil complex. Second, every improvement arrives together with lower average gross exposure (1.5 to 1.8 against baseline’s 2.00) and an improvement in expected shortfall, while maximum drawdown is essentially unchanged at -19.0% ; the de-risking is trimming ordinary variance rather than catching the single worst episode. Third, these are 25 rules applied to the same 234 months, which makes the leaderboard a distribution of outcomes under one design, not 25 independent discoveries.

Table 3. The ten strongest macro overlays on baseline carry, ranked by full-sample gross Sharpe. *Act.* is the share of months in which the trigger fired. Source: 08v6_macro_overlay_performance.parquet.

50% overlay on	Act.	Ret.	Vol.	SR	Δ SR	ES	Gr.
Baseline carry	—	+5.24%	9.07%	+0.58	—	-6.47%	2.00
OIL_return_1m	20%	+5.23%	8.03%	+0.65	+0.073	-5.30%	1.80
COMMODITY_momentum_3m	22%	+5.19%	8.12%	+0.64	+0.061	-5.69%	1.78
MXEF_tail_return_1m	17%	+5.29%	8.30%	+0.64	+0.060	-5.72%	1.83
MXEF_return_1m	17%	+5.29%	8.30%	+0.64	+0.060	-5.72%	1.83
MXEF_momentum_3m	20%	+5.11%	8.03%	+0.64	+0.058	-5.59%	1.80
MXEF_drawdown_6m	23%	+4.97%	7.83%	+0.63	+0.057	-5.24%	1.77
SPX_momentum_3m	20%	+5.07%	8.15%	+0.62	+0.043	-5.74%	1.80
COMMODITY_return_1m	18%	+5.21%	8.38%	+0.62	+0.043	-5.88%	1.82
MOVE_change_1m	18%	+5.02%	8.22%	+0.61	+0.033	-5.62%	1.82
CURVE_2S10S_level	13%	+5.10%	8.56%	+0.60	+0.018	-5.79%	1.87

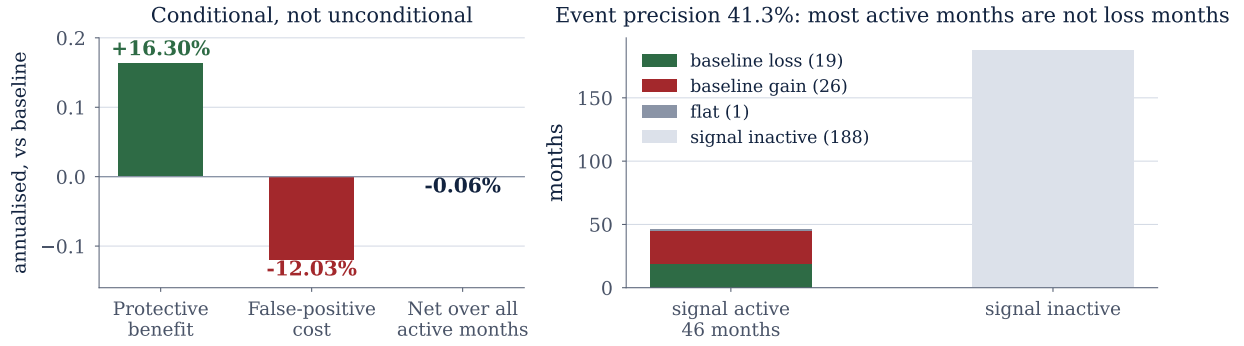


Figure 5. Conditional attribution of the oil overlay. Left: annualised mean relative return against baseline in signal-active baseline-loss months, signal-active baseline-gain months, and across all active months. Right: the composition of the 46 active months.

8.2 The oil case study

The overlay triggered by `OIL_return_1m` is the most instructive single result in the study, and the one most easily misreported. Its full-sample gross Sharpe is 0.651 against baseline’s 0.578, its frozen evaluation Sharpe is 1.59, its average gross exposure is 1.80, and its expected shortfall improves by 1.17 percentage points.

The mechanism is best seen conditionally. Define the *protective benefit* as

$$12 \times \text{mean} [r_t^{\text{overlay}} - r_t^{\text{baseline}} \mid E_{t-1} = 1 \text{ and } r_t^{\text{baseline}} < 0], \quad (9)$$

the annualised average relative return in months where the signal was active *and* the reference strategy subsequently lost money. For the oil overlay this equals +16.30%, from an average monthly relative return of 1.36% across the 19 such months.

What the 16.30% is, and is not. It is a conditional annualised attribution statistic computed over a subset of months selected on both the signal and the realised outcome. It is **not** a 16.3% improvement in annual portfolio return, and it is **not** alpha. Quoting it without its conditioning set, or without the counterpart below, would be misleading.

The counterpart is the false-positive cost, defined identically over signal-active months where the reference *gained*: -12.03% annualised across 26 such months. Event precision — the share of active months that were baseline-loss months — is 41.3%, or 19 of 46. Netting the two effects across all active months leaves an annualised active-period delta of -0.06%, a benefit-to-cost ratio of 0.99.

Table 4. The oil 50% gross-scaling overlay on baseline carry.

Statistic	Value	Reading
Full-sample gross Sharpe	+0.65	baseline 0.58
Frozen-evaluation gross Sharpe	+1.59	baseline over the same 47 months 1.78
Average gross exposure	1.80	baseline 2.00
Expected shortfall (5%)	-5.30%	improvement of +1.17 points
Signal-active months	46 of 234	19.7% activation
Event precision	41.3%	19 loss months of 46 active
Conditional protective benefit	+16.30%	active <i>and</i> baseline-loss months
Conditional false-positive cost	-12.03%	active <i>and</i> baseline-gain months
Net over all active months	-0.06%	the two conditional effects cancel

The full-sample Sharpe still improves, from 0.578 to 0.651. That improvement comes from the denominator: the overlay removes variance and improves the expected shortfall while leaving the long-run return essentially unchanged. The correct characterisation of the oil overlay is therefore *defensive gross management with a favourable variance trade*, not mean-return timing. The distinction is not pedantic — a timing model would be expected to survive a change in the cost of being wrong, and this one would not.

8.3 Active-period attribution

Appendix Figure 12 compares each overlay’s performance during its own signal-active months against baseline over the identical months. The pattern is consistent with the 41.3% precision statistic: active months are, on average, *not* losing months for baseline carry. The overlays run half-sized through a mixture of good and bad months, which is why any protective value must be assessed conditionally rather than by averaging over all active periods.

8.4 Crisis and regime results

Section 10.4 treats crisis attribution in detail. The key point for the standalone overlays is that protection is episode-specific: the oil overlay cuts the Covid drawdown from -7.2% to -1.4% but leaves the global financial crisis untouched, because the oil signal was largely inactive through 2008.

8.5 Composite macro signals

Notebook 08 also tests five pre-defined composite rules: an OR rule active when any of the 25 feature events fires; a two-or-more confirmation rule; a volatility-family confirmation rule; a confirmed-market-stress rule spanning volatility, equity, dollar and rates families; and a continuous rule built on

$$\text{MacroRisk}_t = \frac{1}{25} \sum_{j=1}^{25} \rho_j z_{j,t}, \quad (10)$$

the average risk-direction-normalised z -score, thresholded at its own rolling 80th percentile. Only the last uses signal *strength*; the others count binary events.

The answer to whether breadth beats the best single feature is no. The best composite adds +0.037 Sharpe against baseline, while the best individual features add between +0.06 and +0.07. The reason is that the breadth is *feature-level*, not asset-class-level: four of the 25 features are MXEF transformations and five are volatility indices, so an “any-of-25” rule fires repeatedly on the same underlying state. The OR composite is active 72% of the time and spends most of that de-risking a book that is not in trouble; the confirmation rules require agreement and are consequently late. Describing these composites as diversified across independent signals would misstate what they are.

Table 5. Composite macro rules against baseline carry, full sample.

Composite	Activation rule	Active	Sharpe	ΔSR	Gross
OR	any one of the 25 feature events is active	72%	+0.55	-0.030	1.28
Two-or-more confirmation	at least two of the 25 feature events are active	59%	+0.56	-0.020	1.41
Volatility confirmation	at least one volatility-family event (VIX, JPMVXYEM, MOVE)	46%	+0.61	+0.035	1.54
Confirmed market stress	at least one event among volatility, equity, dollar and rates families	65%	+0.59	+0.011	1.35
Continuous macro-risk rule	$\text{MacroRisk}_t \geq$ its own historical 80th percentile	19%	+0.61	+0.037	1.81
Baseline carry	—	0%	+0.58	—	2.00

Table 6. The three inherited option architectures.

Label	Notebook 07 strategy	Economic role	Gross exposure
A	G10-aware ATM conditioning ($\lambda_{G10}=0.5$, $\lambda_{EM}=0$)	selection on ATM volatility	fixed, ≈ 2.0
B	Butterfly-conditioned carry ($\lambda_B = -0.5$)	selection on smile curvature	fixed, ≈ 2.0
C	Joint option-risk scaling, variable-gross ($\gamma=0.5$)	exposure control, same basket	variable, ≤ 2.0

9 Option-conditioned carry

9.1 Notebook 07 recap

The option layer is inherited from Notebook 07 and is not re-derived here. Notebook 08 reproduces its weights, returns, turnover and costs from the stored weight files before using them, and the reconstruction matches the stored performance outputs exactly. Option signals are monthly within-asset-class cross-sectional standardisations of the one-month ATM implied volatility, the 25-delta depreciation skew and the 25-delta butterfly.

9.2 The A, B and C mechanisms

Architectures **A** and **B** are *selection* mechanisms: they change what the portfolio holds. The ranking score is carry multiplied by a bounded option factor,

$$\text{Score}_{i,t} = C_{i,t} \text{ clip}(1 + \lambda z_{i,t}^X, 0.25, 2.00), \quad (11)$$

so the multiplier is strictly positive, the sign of carry survives, and only the ordering can change. Architecture A sets $\lambda = 0.5$ on ATM volatility for G10 currencies and 0 for EM — asymmetric by design, because the carry $\times$ ATM interaction documented in Notebook 06 was a G10 result rather than a universal one. Architecture B sets $\lambda_B = -0.5$ on the butterfly, so high smile curvature compresses the carry signal and pushes a currency towards the middle of the ranking. Both leave average gross exposure at 2.00.

Architecture **C** is an *exposure control* mechanism: it keeps the baseline basket and shrinks positions that look expensive to hold,

$$m_{i,t} = \text{clip}\left(\frac{1}{1 + \gamma \sum_X \max(z_{i,t}^X, 0)}, 0.25, 1.00\right), \quad \gamma = 0.5, \quad (12)$$

summing over ATM volatility, skew and butterfly. Under the variable-gross convention the scaled legs are renormalised to $G_t = \min(L_t, S_t)$, where L_t and S_t are the scaled long and short gross, so the book remains net-neutral while total gross exposure falls below 2.00.

Table 7. The eight core option architectures, full sample (234 months) with the frozen evaluation Sharpe in the final column. Source: 08v6_core_option_comparison.parquet.

Arch.	Mechanism	Full sample, 234 months								Eval.
		Ret.	Vol.	Sharpe	Δ SR	Max DD	ES	Gross	Turn.	Sharpe
Baseline	baseline carry	+5.24%	9.07%	+0.58	+0.000	-19.00%	-6.47%	2.00	0.18	+1.78
A	selection improvement	+5.70%	8.93%	+0.64	+0.060	-20.81%	-6.14%	2.00	0.23	+1.96
B	selection improvement	+5.26%	7.99%	+0.66	+0.080	-16.89%	-5.14%	2.00	0.29	+1.87
C	variable-gross risk control	+2.81%	4.61%	+0.61	+0.030	-11.80%	-3.08%	1.16	0.18	+1.64
A+B	selection improvement	+5.35%	7.77%	+0.69	+0.110	-17.51%	-4.96%	2.00	0.33	+1.69
A+C	selection + variable-gross risk control	+3.10%	4.43%	+0.70	+0.121	-11.57%	-2.92%	1.14	0.21	+1.94
B+C	selection + variable-gross risk control	+2.70%	4.63%	+0.58	+0.005	-12.65%	-3.06%	1.30	0.25	+1.72
A+B+C	selection + variable-gross risk control	+2.66%	4.51%	+0.59	+0.011	-13.86%	-2.96%	1.29	0.27	+1.49

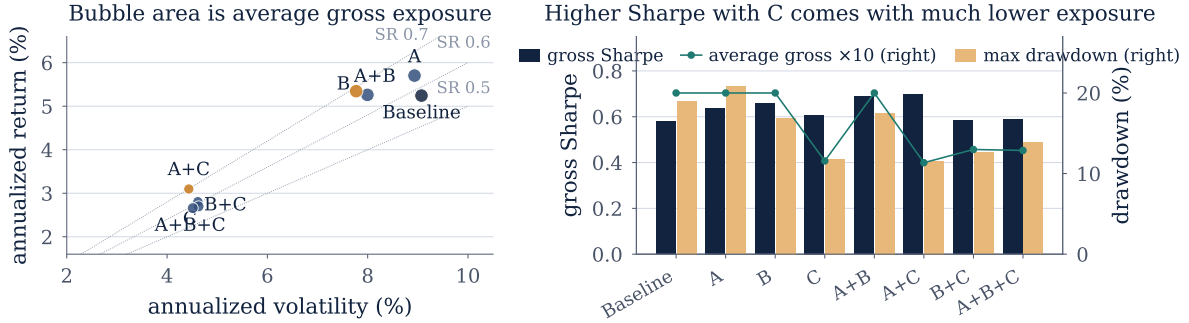


Figure 6. Left: annualised return against volatility, with bubble area proportional to average gross exposure and dotted Sharpe isoquants. Right: Sharpe, maximum drawdown and average gross for the same eight books.

9.3 Core architecture combinations

Table 7 shows two routes to a similar Sharpe. **A+B** reaches 0.688 at unchanged average gross of 2.00, earning 5.35% against baseline’s 5.24% at lower volatility (7.77% against 9.07%). This is a selection result: the same amount of capital is deployed and the composition of the book is doing the work. **A+C** reaches 0.699 — nominally the best of the eight — at average gross of 1.14 and an annualised return of 3.10%, with maximum drawdown improving from -19.0% to -11.6% .

9.4 Selection versus risk-control decomposition

The A+C result requires care. Scaling a portfolio by a constant factor scales return and volatility proportionally and leaves Sharpe unchanged, so the fact that A+C improves Sharpe means its scaling is not constant: it carries information about when to be smaller. But the improvement in absolute drawdown, and part of the improvement in volatility, follow mechanically from running roughly half the exposure. Levering A+C back to gross 2.00 would raise financing and capacity questions the backtest does not address, and would widen the -11.6% drawdown back towards baseline’s.

Figure 6 makes the decomposition visible: every architecture containing C sits in the low-volatility, low-gross cluster, and every fixed-gross architecture sits near baseline on the return–volatility plane. Selection moves the numerator; exposure control moves the denominator. Both are legitimate, and reporting either without average gross would be misleading.

9.5 Evaluation-sample results

In the frozen 2022–2026 window A+B posts 1.69 and A+C posts 1.94. These numbers should never be quoted alone, because *baseline carry itself* posts 1.78 over the same 47 months. Relative to that contemporaneous benchmark, A+C is +0.16 and A+B is -0.09 ; the oil overlay is -0.19 . The evaluation window opens after the 2022 dollar and rates shock and contains no comparable episode, which is a favourable environment for a book long EM against low-yield funders. Full-sample performance is used as the descriptive headline throughout this document, with evaluation

Table 8. Selected combined strategies against their option parents. Source: 08v6_macro_extension_incremental_results.parquet.

Parent	Overlay	Gross Sharpe			vs. parent		Avg. gross
		Par.	Comb.	Δ	Ret.	MDD	
Baseline	OIL	+0.58	+0.65	+0.073	-0.01%	+0.00%	2.00 $\rightarrow$ 1.80
A	MXEF	+0.64	+0.71	+0.071	+0.12%	+0.00%	2.00 $\rightarrow$ 1.83
A	OIL	+0.64	+0.70	+0.060	-0.10%	+0.00%	2.00 $\rightarrow$ 1.80
B	OIL	+0.66	+0.72	+0.060	-0.10%	+0.00%	2.00 $\rightarrow$ 1.80
A+B	OIL	+0.69	+0.74	+0.048	-0.16%	+0.00%	2.00 $\rightarrow$ 1.80
A+C	OIL	+0.70	+0.73	+0.028	-0.17%	+0.00%	1.14 $\rightarrow$ 1.02
A+C	MXEF	+0.70	+0.74	+0.038	-0.07%	+0.00%	1.14 $\rightarrow$ 1.03
A+B+C	OIL	+0.59	+0.58	-0.008	-0.23%	+0.33%	1.29 $\rightarrow$ 1.16

results as robustness evidence.

10 Macro and option strategies combined

10.1 Construction

A combined strategy applies the macro overlay to an option-conditioned parent by multiplying the parent’s own weights, exactly as in Equation (7) with w^{parent} now being the option architecture’s weights. For a fixed-gross parent such as A+B this takes gross from 2.00 to 1.00 on active months. For a variable-gross parent such as A+C it takes gross to half of whatever the parent was already running; it does not force gross to any fixed level. The two layers of de-risking compound.

10.2 Incremental value relative to the parent

The economically meaningful comparison is against the parent, not against baseline. Measured against baseline, almost every combination looks like an improvement — but so does the parent on its own, and attributing the parent’s contribution to the macro layer would be a straightforward error of attribution.

Table 8 shows diminishing returns to stacking exposure rules. Adding the oil overlay to baseline carry is worth +0.073 Sharpe. Adding the same overlay to A+C is worth +0.028, and it cuts average gross from 1.14 to 1.02. The macro layer and the option-risk layer are detecting the same underlying stressed state through different instruments; where they overlap, the second one mostly adds turnover.

10.3 Cross-product analysis

Notebook 08 crosses all eight option parents with all 25 macro features, a pre-defined grid of 200 combinations in which every outcome is reported. This is deliberately not framed as a search for the best of 271 strategies: the question is whether the interaction between option-based portfolio construction and macro risk conditioning has a systematic structure, and the answer is visible in Figure 7.

Two patterns stand out. Columns dominate rows: whether a macro feature helps is mostly a property of the feature, and the left edge of the sorted heatmap is the EM-equity, commodity and oil complex — the same group the regressions flagged. Rows fade downwards: the parents that gain least are those that already contain architecture C and therefore already scale their own exposure.

Of the 200 combinations, **89 improve** their parent’s Sharpe and **111** do not. Macro conditioning on top of option conditioning is therefore selective rather than systematic. Appendix Figure 11 shows the distribution: it is centred slightly below zero, with a median of -0.004 , which

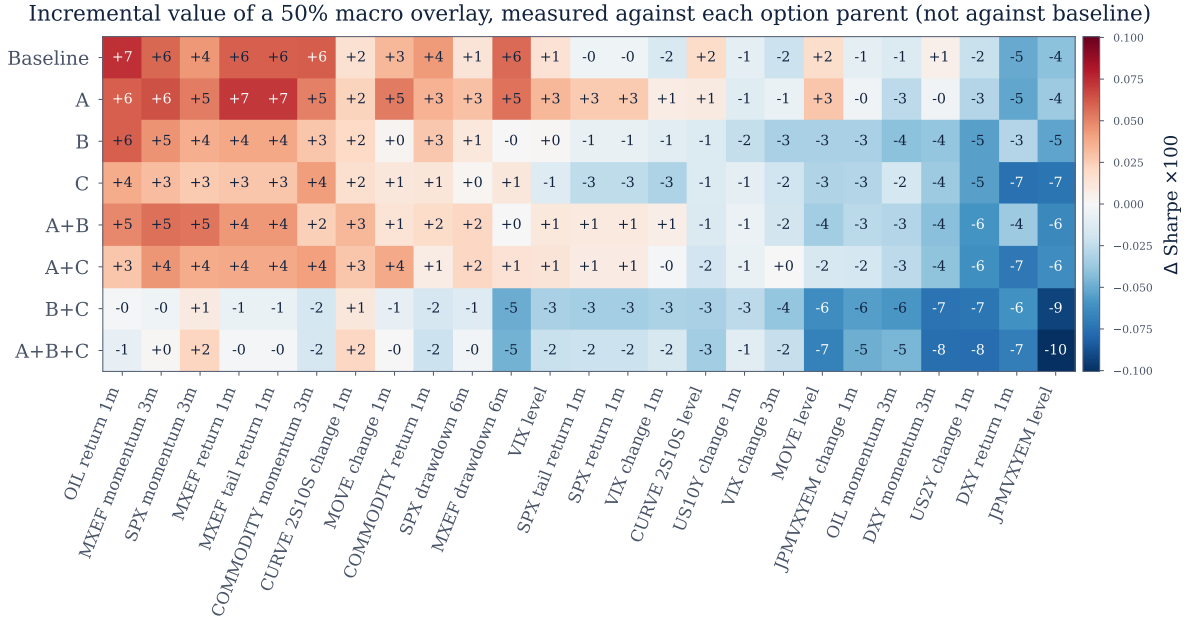


Figure 7. Incremental Sharpe change ($\times 100$) of a 50% macro overlay, measured against each option parent. Columns are sorted by average effect.

is what a genuine but weak and feature-specific effect looks like. A systematic effect would sit clearly to the right of zero.

10.4 Risk-period and crisis performance

Two definitions of a bad month are used, and they are never combined. *Fixed crisis windows* are five calendar episodes selected with hindsight: the global financial crisis, the 2013 taper tantrum, the 2015–16 EM and China stress, the 2020 Covid shock and the 2022 dollar and rates shock. They test whether a strategy survived history. *Rule-defined active periods* are the months in which a given strategy’s own signal fired; they test whether the rule does what it claims. A strategy can perform well across the fixed crises and still fire mostly in calm months, so the two must be reported separately.

Selection architectures do not reliably help in a crisis: A+B lost more than baseline in both the global financial crisis and the taper tantrum. The exposure-based routes do. A+C roughly halves the losses in both the GFC and Covid, and the oil overlay cuts the Covid loss from -7.2% to -1.4% while leaving the GFC essentially untouched — the oil signal was largely inactive in 2008.

The general lesson is that protection is episode-specific. A signal built on one transmission channel protects against shocks that travel through that channel and is blind to the rest. That is an argument for describing precisely what a rule does, not for calling any of these overlays a hedge.

11 Robustness and research governance

11.1 Frozen out-of-sample evidence

The 47-month evaluation window was fixed before the strategies in this document were evaluated on it, and no evidence label or design choice uses it. Its main use is negative: it demonstrates how misleading a raw out-of-sample Sharpe can be when the period itself is favourable. Every strategy examined posts an evaluation Sharpe above 1.4, and baseline carry posts 1.78. Appendix Table 10 reports each strategy alongside its delta against baseline over the identical months, which is the only comparison that removes the period effect.

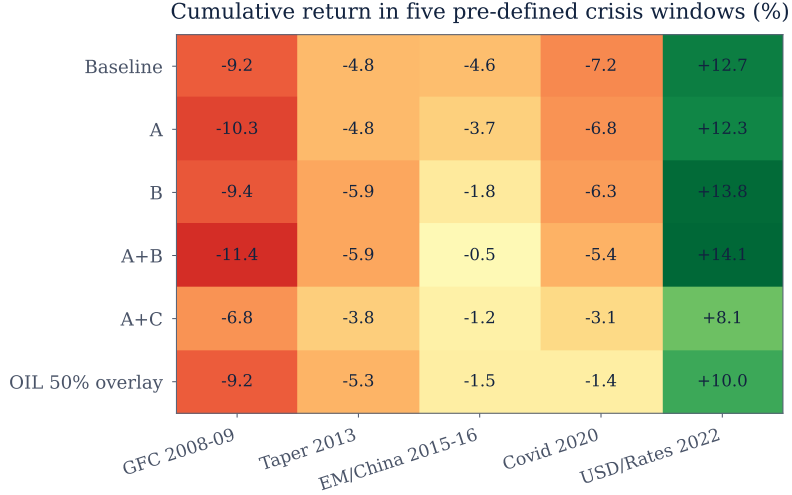


Figure 8. Total return in each fixed crisis window. The underlying table with average gross exposure is in Appendix E.

11.2 Point-in-time and look-ahead auditing

The macro, option and strategy-level lineage audits report **zero** future-horizon violations. All 271 canonical strategies carry a complete point-in-time lineage row. However, the depth of verification is uneven, and the status taxonomy records how far it could go: 120 strategies are verified at calendar-date resolution with intraday timestamps unavailable; 96 have source logic that is documented and inherited but not independently reconstructed here; 50 have source metadata too thin to verify beyond lineage; and 5 have verified return timing and no conditioning signal to audit. Verification evidence is therefore complete for 221 strategies and incomplete for 50. The statement that no leakage was detected is accurate; the statement that all strategies are independently verified would not be.

11.3 Transaction costs

Net returns subtract a roll cost applied to gross notional $\sum_i |w_{i,t}|$ and a rebalancing cost applied to one-way turnover $\frac{1}{2} \sum_i |w_{i,t} - w_{i,t-1}|$, both at a common per-unit rate. Appendix Table 11 reports net Sharpe from 1 to 50 basis points. The ordering of strategies is stable at realistic levels. At higher cost assumptions the selection-based books degrade fastest, because they turn over more (0.33 for A+B against 0.18 for baseline), while gross-controlled books degrade more slowly because both cost terms scale with a smaller book. Cost is part of the mechanism story rather than a separate caveat: the two routes to a similar Sharpe have different cost sensitivities.

11.4 Multiple testing

At the strategy level, 36 of the 270 active non-reference strategies have a raw HAC p -value below 0.05 on their mean active return. After Benjamini–Hochberg correction, **none** has $q < 0.10$.

This is the most important statistical result in the study and it constrains every claim made elsewhere in this document. It is also why the Tier 1 shortlist is empty. Two qualifications belong with it. First, the correction is conservative here because the 270 strategies are far from independent — they share parents, share signals, and in some cases differ only in a scaling constant — so the effective number of tests is well below 270. Second, the correction is applied to the mean active return, which is precisely the quantity the macro overlays are *not* designed to improve; a variance or tail-focused test would ask a different question. Neither qualification rescues a claim of demonstrated profitability.

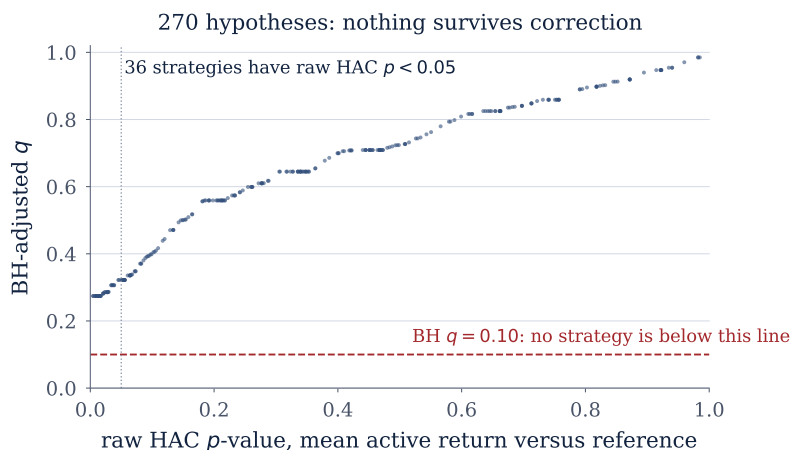


Figure 9. Raw HAC p -values against Benjamini–Hochberg adjusted q -values for the 270 active non-reference strategies.

11.5 De-duplication

The raw strategy catalogue contains 323 named entries, which collapse to 271 canonical return streams after removing 52 aliases with numerically identical weights. Every count in this document uses the canonical figure. Counting aliases would inflate every breadth statistic in the study.

11.6 Strategy role and tier taxonomy

Strategies are classified into a research-confidence taxonomy: Tier 1 (robust shortlist) contains 0 strategies, Tier 2 (promising candidate) 98, Tier 3 (exploratory watchlist) 172, and 1 strategy — baseline carry — is held as a reference benchmark and not ranked. Tier 2 is not a count of 98 validated profitable strategies; it means 98 return streams cleared several screens and none cleared all of them. The empty Tier 1 is the intended output of a strict screen rather than a failure of the analysis, and reporting it is more informative than relaxing the screen until something passes.

12 Conclusions

- 1. Macro predictability is real, and it is about risk.** Across 25 point-in-time features, eight carry supported evidence on severe-loss probability, four on return magnitude, and three on mean returns. The obsolete summary that no macro feature predicts mean returns is wrong. The accurate summary is that macro predictive evidence is strongest and broadest for carry tail risk, with additional evidence for risk magnitude and more limited but non-zero evidence for mean returns.
- 2. That makes macro information a sizing tool rather than a timing tool.** The oil overlay produces a large conditional benefit in signal-active baseline-loss months (+16.30% annualised conditional), most of which is paid back through false positives (−12.03%), netting −0.06% across all active months at 41.3% precision. Its full-sample Sharpe improvement from 0.578 to 0.651 comes from variance reduction, not from added return. Defensive gross management is the honest description.
- 3. Option information improves carry through two distinct channels.** A+B improves Sharpe to 0.688 at unchanged gross 2.00 (selection); A+C reaches 0.699 at gross 1.14 with a −11.6% drawdown (exposure control). Both are useful, and conflating them credits de-risking as if it were selection skill.
- 4. Macro adds selectively after option conditioning, not systematically.** Of 200 parent-feature combinations, 89 improve the parent and 111 do not, and the improvements concentrate in the same EM-equity, commodity and oil features the regressions flagged in advance. Parents that already scale their own exposure gain the least. No claim is made that macro

conditioning improves every option-conditioned strategy.

The governing caveat. No strategy in this study survives multiple-testing correction on its mean active return, and the Tier 1 shortlist is empty by construction. What the evidence supports is a claim about mechanism: which moment of the carry return distribution macro information predicts, and which portfolio action that justifies. It does not support a claim of a validated tradable edge.

13 Limitations

Statistical. No strategy survives Benjamini–Hochberg correction. The frozen evaluation window is 47 months, too short to resolve any Sharpe difference reported here. Univariate regressions identify association, not causation, and no causal interpretation is offered. The 25 macro features are strongly cross-correlated, so the counts of supported features overstate the number of independent findings.

Construction. Two of the 25 features are numerically duplicated under different labels. The 50% scaling multiplier and the 80th/20th percentile thresholds are fixed by judgement rather than estimated: this avoids in-sample fitting but leaves the rules un-optimised, and the results should not be read as an upper bound on what a tuned version could achieve. Composite breadth is feature-level rather than asset-class-level.

Verification. Fifty strategies have source information too thin to verify beyond lineage, and the four combined option architectures inherit selection logic that is documented but not independently re-derived in this work.

Economic. Sharpe ratios throughout are gross: annualised arithmetic return divided by annualised volatility, with no risk-free rate subtracted and no financing cost applied. Transaction costs are a linear per-unit charge on roll notional and turnover, with no market impact and no capacity constraint. Variable-gross results are not leverage-matched to baseline, so their drawdown and volatility advantages are not directly comparable to fixed-gross alternatives at equal risk.

A The 25 macro features

Table 9. All 25 formation-time macro features. Risk sign +1 means higher is riskier. Support flags: M mean, G magnitude, T tail; a dot means not supported.

Series	Category	Feature id	Transformation	Risk sign	Support
VIX	volatility	VIX_level	level at formation month end	+1	.G.
VIX	volatility	VIX_change_1m	1-month difference	+1	...
VIX	volatility	VIX_change_3m	3-month difference	+1	...
JPMVXYEM	volatility	JPMVXYEM_level	level at formation month end	+1	...
JPMVXYEM	volatility	JPMVXYEM_change_1m	1-month difference	+1	.GT
MOVE	rates volatility	MOVE_level	level at formation month end	+1	...
MOVE	rates volatility	MOVE_change_1m	1-month difference	+1	...
DXY	dollar	DXY_return_1m	1-month log return	+1	...
DXY	dollar	DXY_momentum_3m	3-month log return	+1	...
SPX	equity	SPX_return_1m	1-month log return	-1	...
SPX	equity	SPX_momentum_3m	3-month log return	-1	...
SPX	equity	SPX_drawdown_6m	price / 6-month rolling max - 1	-1	...
SPX	equity	SPX_tail_return_1m	1-month log return (lower-tail label)	-1	...
MXEF	EM equity	MXEF_return_1m	1-month log return	-1	M.T
MXEF	EM equity	MXEF_momentum_3m	3-month log return	-1	.GT
MXEF	EM equity	MXEF_drawdown_6m	price / 6-month rolling max - 1	-1	.GT
MXEF	EM equity	MXEF_tail_return_1m	1-month log return (lower-tail label)	-1	M.T
US2Y	rates	US2Y_change_1m	1-month difference	+1	...
US10Y	rates	US10Y_change_1m	1-month difference	+1	...
CURVE	rates	CURVE_2S10S_level	level at formation month end	+1	...
CURVE	rates	CURVE_2S10S_change_1m	1-month difference	+1	...
COMMODITY	commodity	COMMODITY_return_1m	1-month log return	-1	..T
COMMODITY	commodity	COMMODITY_momentum_3m	3-month log return	-1	M.T
OIL	oil	OIL_return_1m	1-month log return	-1	..T
OIL	oil	OIL_momentum_3m	3-month log return	-1	...

B Metric definitions

$$\begin{aligned}
\text{Gross exposure} &= \sum_i |w_{i,t}| \\
\text{Annualised arithmetic return} &= 12 \times \text{mean}(r_t) \\
\text{Annualised volatility} &= \sqrt{12} \times \text{sd}(r_t) \\
\text{Gross Sharpe} &= \frac{\text{annualised arithmetic return}}{\text{annualised volatility}} \\
\text{Annualised long-leg contribution} &= 12 \times \text{mean} \left(\sum_{i: w_{i,t} > 0} w_{i,t} R_{i,t+1} \right) \\
\text{Annualised short-leg contribution} &= 12 \times \text{mean} \left(\sum_{i: w_{i,t} < 0} w_{i,t} R_{i,t+1} \right) \\
\text{One-way turnover} &= \frac{1}{2} \sum_i |w_{i,t} - w_{i,t-1}| \\
\text{Protective benefit} &= 12 \times \text{mean} \left[r_t^{\text{strategy}} - r_t^{\text{reference}} \mid \text{signal active, } r_t^{\text{reference}} < 0 \right]
\end{aligned}$$

The Sharpe ratio used throughout is a *gross* Sharpe: no risk-free rate is subtracted, and it should not be described as an excess-return Sharpe. The protective benefit is a conditional annualised attribution statistic computed over a subset of months selected on both the signal state and the realised sign of the reference return; it is not a portfolio return.

C Frozen-evaluation results

Table 10. Frozen evaluation window, 2022-08-31 to 2026-06-30, 47 months. The delta column compares against baseline carry over the same months.

Strategy	Ret.	Vol.	Sharpe	Δ SR vs base	Max DD	Gross	Tier
Baseline	+10.95%	6.16%	+1.78	+0.000	-6.61%	2.00	Reference / not ranked
A	+12.24%	6.25%	+1.96	+0.180	-6.85%	2.00	Tier 2 promising candidate
B	+10.59%	5.66%	+1.87	+0.090	-6.36%	2.00	Tier 2 promising candidate
C	+5.69%	3.47%	+1.64	-0.140	-3.65%	1.15	Tier 3 exploratory watchlist
A+B	+9.85%	5.81%	+1.69	-0.085	-6.08%	2.00	Tier 3 exploratory watchlist
A+C	+6.56%	3.38%	+1.94	+0.161	-3.85%	1.13	Tier 2 promising candidate
OIL 50\% overlay	+8.93%	5.62%	+1.59	-0.190	-6.61%	1.79	Tier 3 exploratory watchlist

D Transaction-cost sensitivity

Table 11. Net gross-Sharpe under a linear roll-plus-rebalancing cost at per-unit rates from 1 to 50 basis points.

Strategy	Gross	1 bp	2 bp	5 bp	10 bp	25 bp	50 bp
Baseline	+0.58	+0.55	+0.52	+0.43	+0.29	-0.14	-0.86
A	+0.64	+0.61	+0.58	+0.49	+0.34	-0.11	-0.86
B	+0.66	+0.62	+0.59	+0.49	+0.31	-0.20	-1.06
A+B	+0.69	+0.65	+0.62	+0.51	+0.33	-0.21	-1.11
A+C	+0.70	+0.66	+0.63	+0.52	+0.33	-0.21	-1.13
OIL 50\% overlay	+0.65	+0.62	+0.59	+0.50	+0.34	-0.13	-0.90

E Fixed crisis windows

Table 12. Total return in each fixed crisis window, and average gross exposure across all five.

Strategy	GFC 2008-09	Taper 2013	EM/China 2015-16	Covid 2020	USD/Rates 2022	Avg. gross
Baseline	-9.18%	-4.85%	-4.64%	-7.18%	+12.73%	2.00
A	-10.29%	-4.85%	-3.74%	-6.79%	+12.32%	2.00
B	-9.37%	-5.88%	-1.76%	-6.29%	+13.77%	2.00
A+B	-11.38%	-5.88%	-0.45%	-5.37%	+14.09%	2.00
A+C	-6.80%	-3.84%	-1.23%	-3.10%	+8.08%	1.11
OIL 50\% overlay	-9.18%	-5.26%	-1.51%	-1.45%	+9.98%	1.66

F Point-in-time status and strategy taxonomy

Table 13. Point-in-time verification status across the 271 canonical strategies.

Point-in-time status	Strat.	What it means
calendar date verified, intraday unverified	120	signal dates precede the return window; intraday timestamps unavailable
source logic supported, not independently rebuilt	96	construction logic documented and inherited, selection not re-derived here
insufficient source information	50	source metadata too thin to verify beyond lineage
return timing verified, no conditioning signal	5	the strategy has no conditioning signal to audit
Total	271	complete lineage for all; verification evidence complete for 221, incomplete for 50

Table 14. Canonical strategy families after de-duplication.

Strategy family	Canonical strategies
macro extension	161
standalone macro	46
ultimate exploratory matrix	35
parent-relative tilt	16
core option	8
composite macro	5
Total	271

Table 15. Research-confidence taxonomy.

Research-confidence tier	Count	Reading
Tier 1 robust shortlist	0	passes every screen simultaneously
Tier 2 promising candidate	98	survives several screens; not independently validated
Tier 3 exploratory watchlist	172	retained for the record only
Reference / not ranked	1	Baseline carry, the comparison benchmark

G Supporting figures

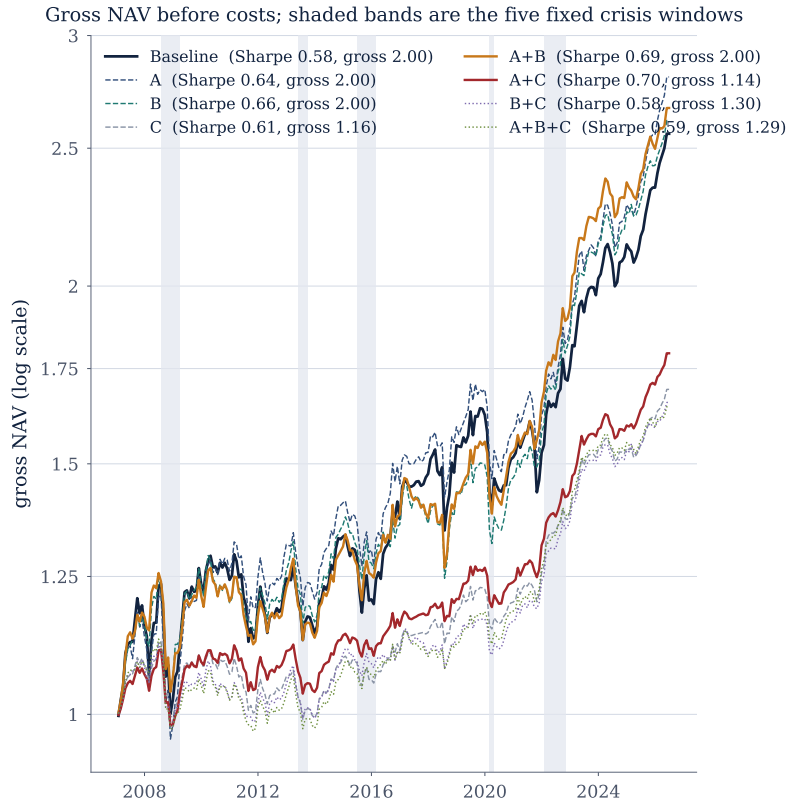


Figure 10. Gross NAV of the eight core option architectures, log scale, with the five fixed crisis windows shaded. Streams are rebuilt from the Notebook 07 and Notebook 08 weight files and reproduce 08v6_core_option_comparison.parquet exactly.

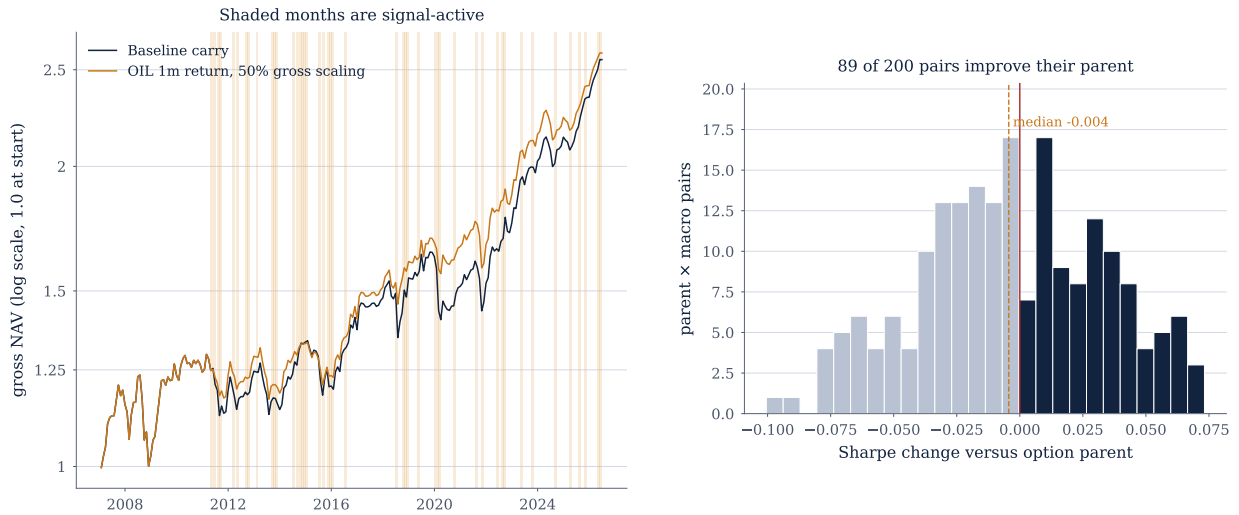


Figure 11. Left: gross NAV of baseline carry and the oil 50% overlay, with signal-active months shaded. Right: the distribution of Sharpe changes across the 200 parent-feature combinations; dark bars are the 89 improvements.

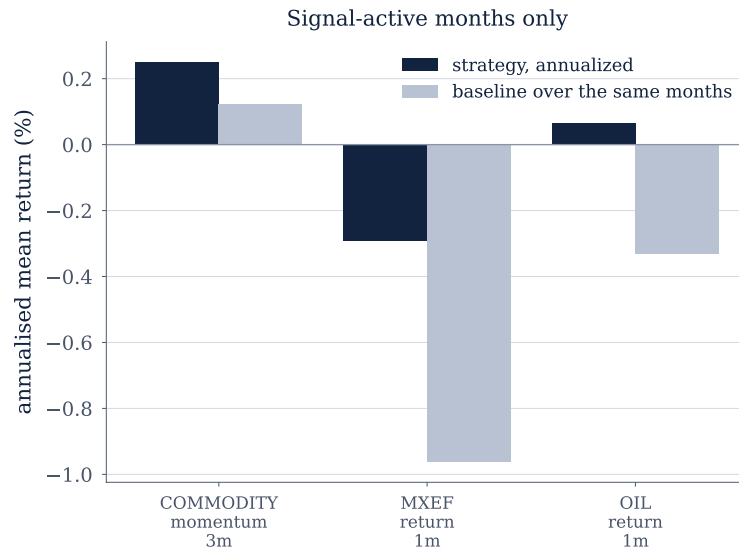


Figure 12. Performance during each strategy’s own rule-defined signal-active months, against baseline carry over the identical months. Source: 08v6_risk_period_results.parquet.

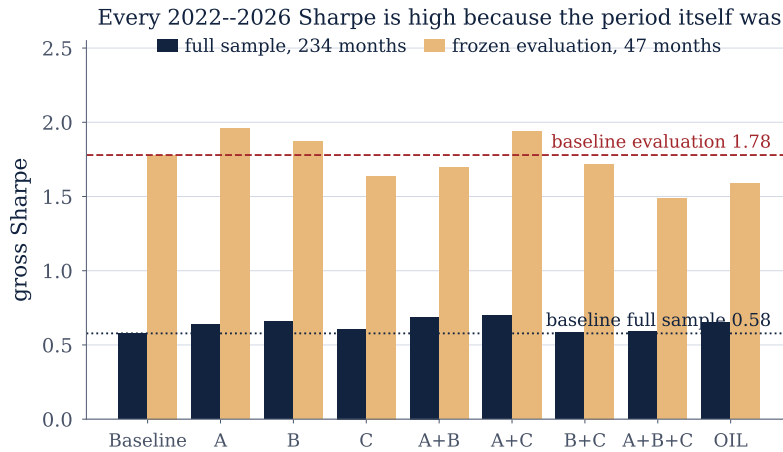


Figure 13. Full-sample against frozen-evaluation gross Sharpe. The evaluation bars are high for every strategy because baseline carry itself posts 1.78 over the same 47 months.

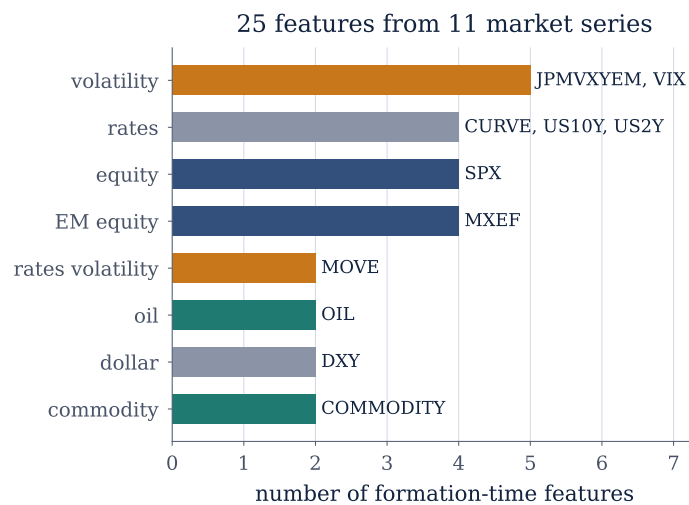


Figure 14. Number of formation-time features derived from each market series, coloured by economic category.

H Fact-check register

Every quantitative claim in the presentation and in this document is generated by `scripts/build_assets.py`, which re-derives each number from the Notebook 08 outputs and terminates the build if any check fails. The register below lists each claim with its source object; the machine-readable version is `build/presentation_fact_check_table.csv` and the full extracted fact set is `build/facts.json`.

Claim	Value used	Notebook source object	Source row	Ver.
Full sample 2007-01-31 to 2026-06-30, 234 months	2007-01-31 to 2026-06-30, 234 months	08v6_summary.json	full_sample_dates / full_sample_months	yes
Discovery 2007-01-31 to 2018-08-31, 140 months	['2007-01-31', '2018-08-31'], 140	08v6_summary.json	discovery_dates / discovery_months	yes
Validation 2018-09-30 to 2022-07-31, 47 months	['2018-09-30', '2022-07-31'], 47	08v6_summary.json	validation_dates / validation_months	yes
Evaluation 2022-08-31 to 2026-06-30, 47 months	['2022-08-31', '2026-06-30'], 47	08v6_summary.json	evaluation_dates / evaluation_months	yes
25 point-in-time macro features	25	08v6_macro_feature_audit.parquet	row count	yes
3 supported mean-return features	MXEF_return_1m, MXEF_tail_return_1m, COMMODITY_momentum_3m	08v6_summary.json	supported_mean_features	yes
4 supported magnitude features	VIX_level, JPMVXYEM_change_1m, MXEF_momentum_3m, MXEF_drawdown_6m	08v6_summary.json	supported_magnitude_features	yes
8 supported tail-risk features	JPMVXYEM_change_1m, MXEF_return_1m, MXEF_momentum_3m, MXEF_drawdown_6m, MXEF_tail_return_1m, COMMODITY_return_1m, COMMODITY_momentum_3m, OIL_return_1m	08v6_summary.json	supported_tail_features	yes
14 explicit predictive/contemporaneous matched pairs	14	08v6_summary.json	lead_concurrent_explicit_matched_pair_count	yes
Baseline full-sample Sharpe 0.58, max drawdown -19.0%	0.5779, -0.1900	08v6_core_option_comparison.parquet	Baseline carry	yes
A+B full-sample Sharpe 0.69, average gross 2.0	0.6880, 2.0000	08v6_core_option_comparison.parquet	Core A+B	yes
A+C full-sample Sharpe 0.70, average gross 1.14, MDD -11.6%	0.6989, 1.1351, -0.1157	08v6_core_option_comparison.parquet	Core A+C	yes
Baseline frozen-evaluation Sharpe 1.78	1.7793	08v6_evaluation_leaderboard.parquet	Baseline carry	yes
A+B evaluation Sharpe 1.69, i.e. -0.09 versus evaluation baseline	1.6942, delta -0.0851	08v6_evaluation_leaderboard.parquet	Core A+B	yes
A+C evaluation Sharpe 1.94, i.e. +0.16 versus evaluation baseline	1.9399, delta +0.1606	08v6_evaluation_leaderboard.parquet	Core A+C	yes
OIL 50% overlay full-sample Sharpe 0.65	0.6511	08v6_summary.json	oil_50pct_overlay_summary.full_sample_sharpe	yes
OIL 50% overlay frozen-evaluation Sharpe 1.59	1.5896	08v6_summary.json	oil_50pct_overlay_summary.frozen_evaluation_sharpe	yes
OIL conditional protective benefit 16.3% annualized	16.30%	08v6_summary.json	oil_50pct_overlay_summary.protective_benefit	yes
OIL false-positive cost 12.03% annualized conditional	12.03%	08v6_summary.json	oil_50pct_overlay_summary.false_positive_cost	yes
OIL net active annualized delta -0.06%	-0.06%	08v6_summary.json	oil_50pct_overlay_summary.net_active_annualized_delta	yes
OIL event precision 41.3%	41.30%	08v6_summary.json	oil_50pct_overlay_summary.event_precision	yes
271 canonical strategies	271	08v6_summary.json	canonical_strategy_count	yes
270 active non-reference multiple-testing hypotheses	270	08v6_summary.json	active_non_reference_multiple_testing_count	yes
Tier 1 = 0, Tier 2 = 98, Tier 3 = 172, Reference = 1	0, 98, 172, 1	08v6_summary.json + 08v6_full_strategy_classification_registry.parquet	robust_shortlist_rows / promising_candidate_list_rows / exploratory_watchlist_rows	yes

Claim	Value used	Notebook source object	Source row	Ver.
PIT status: 120 / 96 / 50 / 5	calendar_date_verified_ intraday_unverified=120, source_logic_supported_ but_not_independently_ reconstructed=96, insufficient_source_ information=50, return_ timing_verified_no_ conditioning_signal=5	08v6_summary.json	strategy_point_in_ time_status_counts	yes
All 271 strategies have complete PIT lineage; evidence complete for 221, incomplete for 50	lineage true=271, evidence true=221, false=50	08v6_summary.json	strategy_pit_lineage_ rows_complete_counts / ..._evidence_complete_ counts	yes
Zero future-horizon leakage detected	0	08v6_summary.json	lookahead_violation_ count / macro_pit_ reconciliation_future_ horizon_violations	yes
200 parent x macro combinations: 89 raise and 111 lower parent Sharpe	200 rows, 89 positive, 111 negative	08v6_macro_extension_ incremental_results.parquet	Delta Sharpe versus parent	yes
OIL overlay on baseline adds +0.073 Sharpe; on A+C it adds +0.028	baseline +0.073, A+C +0.028	08v6_macro_extension_ incremental_results.parquet	Baseline carry / Core A+C x OIL_return_1m	yes
A+C + OIL cuts average gross from 1.14 to 1.02	1.1351 to 1.0224	08v6_macro_extension_ incremental_results.parquet	Core A+C x OIL_return_1m average gross	yes
No strategy's active return survives BH correction (0 of 270 with BH q<0.10)	BH<0.10: 0; raw HAC p<0.05: 36	08v6_gross_statistical_tests.parquet	BH_adjusted_p_value	yes